Detector-local bias stacking: even-sample medians

wenu-imaging implementation guide

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1 Scientific contract

For detector d and active pixel p, let $x_{d,p,i}$ be the scan-corrected value in accepted input frame i. Existing frame-mean rejection selects the accepted set A_d independently for each detector. This change does not alter A_d, scan correction, masks, or raw-pair read-noise measurements. Let $n_d = |A_d|$, and write the ordered retained samples at this pixel as $x_{(1)} \le \cdots \le x_{(n_d)}$. The frame index runs across retained exposures at the same pixel, not across pixels or detectors. The master value is

$$B_{d,p} = \begin{cases} x_{((n_d+1)/2)}, & n_d \text{ odd}, \\ (x_{(n_d/2)} + x_{(n_d/2+1)})/2, & n_d \text{ even, average}, \\ x_{(n_d/2)}, & n_d \text{ even, lower}. \end{cases}$$

The processing option `median_even_handling` defaults to `average`; `lower` explicitly retains the historical bias convention. For ten symmetric samples $[-5, -4, -3, -2, -1, 1, 2, 3, 4, 5]$, the results are respectively 0 and -1. No mosaic-wide offset is estimated or removed. Detector-dependent positive structure is not explained away by this change.

2 Numerical implementation and efficiency

`median_pixelvalue_opts` in `src/math/median.c` performs one in-place Wirth order-statistic selection. For even average mode it selects the upper middle value $u = x_{(n_d/2+1)}$. The maximum of the first $n_d/2$ partition entries is then the lower middle value $\ell = x_{(n_d/2)}$. The stored result is

$$\text{pixelvalue}(0.5\,\text{double}(\ell) + 0.5\,\text{double}(u))\,.$$

This avoids overflow from a working-precision sum and rounds once to the working pixel type. Inputs to the helper must be finite, with positive count and a valid enum. Odd counts retain the existing optimized selection helpers. `cube_avg_median_opts` reuses a single timeline buffer across all pixels. A ten-frame float32 timeline occupies 40 bytes; there is no allocation per pixel, full sort, second selection, or additional image-sized scratch. The extra partition scan is linear in sample count; Wirth selection retains its expected-linear cost. Generic Eclipse median wrappers preserve their previous conventions.

3 Configuration, QC and reproducibility

The recipe accepts `[bias] median_even_handling`. Shared processing TOML also accepts `MEF.BIAS.MEDIAN.EVEN.HANDLING`; the typed domain path is `processing.bias_median_even_handling`. An explicit conflict fails before processing. The optional legacy PAF key has the same native name. The effective choice travels through immutable configuration, bias settings, the bias result, and native product serialization. Each detector records it in ASDF characterization

and the FITS card `ESO MPP QC MED EVEN`, including a defaulted choice. Old ASDF products without this optional field remain valid. Master statistics and high-offset diagnostics are computed from that individual detector's resulting pixels, with their existing finite/BPM population definitions. A spatial master median is a separate operation from the temporal median above. Raw/input measurements, accepted membership, raw RON, MAD RON and clipped RON are unchanged for identical inputs and policies.

4 Verification and limits

The native probe checks 3200 independent sorted-oracle cases, odd/even cubes, input preservation, finite extremes, unchanged generic helpers and the actual bias kernel. Actual service profiles cover the effective mode and FITS QC. The Mac contract additionally compares explicit lower against the retained historical adapter, and default/explicit average against independently ordered scan-corrected planes. An independent ASDF reader checks schema, mapped FITS QC, mode provenance, recomputed spatial master statistics and unchanged input QC/RON. Invalid and conflicting settings must fail. The detector-sized timing is an agent-workspace microbenchmark, not a Mac performance qualification. Native Mac synthetic and real-product checks have passed: all 194346 mapped FITS/ASDF cards and independent spatial master statistics agree. The 27 even stacks shift upward by 0.239002–0.312177 ADU; the five odd stacks retain exact pixel bytes. Input QC, frame selection and RON remain unchanged. The owner accepted the change. Native producer revision 74327c4 remains the qualified binary; later test/tool/documentation changes require no relink. Preserve all qualification and product evidence. CCD68/90 positive residuals and inherited zero-width strict frame-rejection behavior are separate from the median choice. The owner decision of 8 November 2026 (issue #289) preserves zero-spread failure and adds an explicit diagnostic before stacking: detector ID, every input filename and its computed corrected mean in ADU. Exactly zero is tested in double precision; positive spread keeps the strict endpoints without a near-zero tolerance. No master is transferred for the failed detector. Actual Mac qualification of this diagnostic change is pending; earlier median acceptance remains valid for its recorded producer.